

Psychoacoustics, pitch, harmony & rhythm

Musical coordinates and uncertain estimates layered onto the acoustic signal.

Psychoacoustic coordinates and MFCCs

A common mel mapping is $m(f) = 2595 \log_{10}(1 + f/700)$. Mel filters aggregate spectral power, producing $E_b = \sum_k M_{b,k} P_k$. MFCCs apply a discrete cosine transform to log filter energies: $c_r = \sum_b \log(E_b + \epsilon) \cos \left[\frac{\pi r}{B} \left(b + \frac{1}{2} \right) \right]$. DCT normalization, number of filters, frequency range and retained coefficients must be stated. MFCCs summarize spectral envelope; they are not direct measurements of instruments or emotions. Loudness, masking and pitch perception require more specific models; RMS is only an acoustic proxy for perceived loudness.

Pitch and chroma

In twelve-tone equal temperament with A4 at 440 Hz, $f(m) = 440 \cdot 2^{(m-69)/12}$, $m(f) = 69 + 12 \log_2(f/440)$. Pitch class is approximately $m \bmod 12$. Accumulate pitch-weighted spectral evidence into $\mathbf{c}_t \in \mathbb{R}^{12}$, with a documented tuning and normalization policy. Harmonics and percussion can contaminate chroma; octave collapse discards register information.

Pitch candidates can come from autocorrelation, cepstral analysis or harmonic-product methods. Polyphonic melody and multiple fundamental-frequency estimation are harder extensions. State where a method assumes monophony.

Harmony as probabilistic inference

A chord-template score is $s(C, t) = \frac{\mathbf{c}_t^\top \mathbf{v}_C}{\|\mathbf{c}_t\| \|\mathbf{v}_C\|}$. This is a similarity score, not a calibrated chord probability. Normalize and calibrate a probabilistic model before displaying $P(C \mid \mathbf{c}_t)$. Temporal transition matrices or HMMs can favor coherent progressions while retaining alternative chord candidates.

Key profiles, pitch-class entropy and harmonic change are additional descriptors. None uniquely determines harmonic tension. Compare explicit definitions such as distance from a tonic profile with listener annotations before interpreting them perceptually.

Circular statistics may describe pitch-class organization, but ordinary chromatic order differs from circle-of-fifths geometry. Choose the coordinate system according to the musical question.

Onsets, tempo and beats

An onset-strength curve o_t summarizes local increases in spectral energy. Its lagged autocorrelation is $R(\ell) = \sum_t o_t o_{t+\ell}$. A peak at lag ℓ frames implies a tempo candidate $60f_s/(H\ell)$ BPM.

Tempograms expose how periodicity varies over time.

Half-time and double-time ambiguity is central: strong evidence for 120 BPM may also support 60 or 240. Display ranked candidates; call them probabilities only after defining and validating a likelihood and prior. Beat tracking adds phase and continuity, while meter estimation adds grouping.

Inter-onset intervals, onset density and timing variability are measurable. Syncopation needs a meter-aware model; counting irregular intervals alone is insufficient. Rhythmic entropy requires a specified event representation and binning scheme.

Multiscale synchronization

Sample-level, frame-level, beat-level and section-level features have different meanings. Map all timestamps to seconds and offer beat-synchronous aggregation for structural comparison. Show aggregation rules and uncertainty in detected beats. Never silently equate one analysis window with one musical event.

Signal mathematics · Statistical structure